WORKED SOLUTION

STEP II 2001, Question 2

Pure • Integer Part Sums

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q2

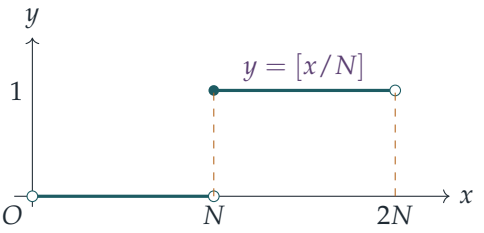
The key idea

The awkward-looking exponent $[k/N]$ takes only three values as k runs from 1 to $2N$, and the sketch shows exactly where it jumps. Once we know that, the sign $(-1)^{[k/N]}$ is constant on each of three blocks of k, so each sum splits into an arithmetic series (part (i)) or a geometric series (part (ii)) on each block. The only care needed is at the endpoints of the blocks.

Solution

The sketch

For $0 < x < N$ we have $0 < x/N < 1$, so $[x/N] = 0$. For $N \leq x < 2N$ we have $1 \leq x/N < 2$, so $[x/N] = 1$. The graph is a step: the jump happens at $x = N$, where the function already takes the value 1, and the right-hand end $x = 2N$ is not included (there the value would be 2).



Filled dots are included and open dots are excluded.

For the sums we need the integer values $k = 1, 2, \ldots, 2N$. Reading off the graph (and

adding the endpoint $k = 2N$):

$$[k/N] = \begin{cases} 0, & 1 \le k \le N-1, \\ 1, & N \le k \le 2N-1, \\ 2, & k = 2N. \end{cases}$$

So $(-1)^{[k/N]}$ is $+1$ for $k \le N-1$, -1 for $N \le k \le 2N-1$, and $+1$ again for the single term $k = 2N$. (When $N = 1$ the first block is empty; every formula below still holds, since an empty sum is 0.)

> **Ask yourself** Why can't we just say the sign is $+$ for the first N terms and $-$ for the next N?

Because the jumps sit exactly at multiples of N: the term $k = N$ already has $[k/N] = 1$, and the last term $k = 2N$ has $[k/N] = 2$, so it is positive. Getting these two endpoints right is the whole question.

Part (i)

Splitting into the three blocks,

$$\sum_{k=1}^{2N} (-1)^{[k/N]} k = \sum_{k=1}^{N-1} k \; - \sum_{k=N}^{2N-1} k \; + \; 2N.$$

The first sum is $\frac{1}{2}(N-1)N$. The second has N terms, from N up to $2N-1$, so it equals N times the average of its first and last terms: $\frac{1}{2}N(3N-1)$. Hence

$$\begin{aligned} \sum_{k=1}^{2N} (-1)^{[k/N]} k &= \tfrac{1}{2}(N^2 - N) - \tfrac{1}{2}(3N^2 - N) + 2N \\ &= -N^2 + 2N = 2N - N^2, \end{aligned}$$

as required.

Part (ii)

The same split gives

$$S_N = \sum_{k=1}^{N-1} 2^{-k} \; - \sum_{k=N}^{2N-1} 2^{-k} \; + \; 2^{-2N}.$$

Both sums are geometric with ratio $\frac{1}{2}$. Using $\sum_{k=a}^{b} 2^{-k} = 2^{-a}\left(1 - 2^{-(b-a+1)}\right) \big/ \frac{1}{2} = 2^{1-a} - 2^{-b}$,

$$\begin{aligned} \sum_{k=1}^{N-1} 2^{-k} &= 1 - 2^{1-N}, \\ \sum_{k=N}^{2N-1} 2^{-k} &= 2^{1-N} - 2^{1-2N}. \end{aligned}$$

Therefore

$$\begin{aligned} S_N &= 1 - 2^{1-N} - 2^{1-N} + 2^{1-2N} + 2^{-2N} \\ &= 1 - 4 \cdot 2^{-N} + 3 \cdot 2^{-2N}. \end{aligned}$$

Writing $t = 2^{-N}$, this is $1 - 4t + 3t^2$, which factorises neatly:

$$\boxed{S_N = 1 - \frac{4}{2^N} + \frac{3}{4^N} = \left(1 - \frac{1}{2^N}\right)\left(1 - \frac{3}{2^N}\right).}$$

As $N \to \infty$, $2^{-N} \to 0$ and $4^{-N} \to 0$, so

$$\lim_{N \to \infty} S_N = 1.$$

This makes sense: as N grows, the positive block $\frac{1}{2} + \frac{1}{4} + \cdots$ gets longer and its sum approaches 1, while the negative block starts so far along that its total, roughly 2^{1-N}, is negligible.

Checks

- $N = 1$: the terms are $k = 1$ with $[1] = 1$ and $k = 2$ with $[2] = 2$. Part (i) gives $-1 + 2 = 1 = 2 - 1$, and part (ii) gives $-\frac{1}{2} + \frac{1}{4} = -\frac{1}{4}$, matching $(1 - \frac{1}{2})(1 - \frac{3}{2}) = -\frac{1}{4}$.
- $N = 2$: the signs are $+, -, -, +$ for $k = 1, 2, 3, 4$. Then $1 - 2 - 3 + 4 = 0 = 4 - 4$, and $\frac{1}{2} - \frac{1}{4} - \frac{1}{8} + \frac{1}{16} = \frac{3}{16} = \frac{3}{4} \cdot \frac{1}{4}$.
- We checked both formulas exactly, with fractions, for every N from 1 to 59. For example $S_{50} \approx 0.999\,999\,999\,999\,996$.

Common slips

- Putting $k = N$ in the positive block, or forgetting that the final term $k = 2N$ has exponent 2 and so is positive.
- Miscounting the terms in $\sum_{k=N}^{2N-1}$: there are N of them, not $N - 1$.

Beyond STEP

The integer-part (floor) function turns up all over number theory: the classic example is Legendre's formula, which says the power of a prime p dividing $n!$ is $\sum_{j \geq 1} [n / p^j]$. Sums like part (ii), where the sign pattern changes in blocks, are also a gentle introduction to rearranging and grouping the terms of series, which analysis courses make rigorous. Davenport's *The Higher Arithmetic* is a lovely place to see the floor function at work.

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